Laporan Tugas Kecil 2 Strategi Algoritma : *Voxelization* Objek 3D menggunakan Octree

1. Identitas Diri
    a. Nama : Muhammad Fakhry Zaki
    b. NIM : 13524087
    c. Kelas : K2

2. Deskripsi Tugas

    Voxelization merupakan proses konversi objek tiga dimensi yang umum menjadi sebuah model yang tersusun dari kubus-kubus kecil seperti di Minecraft. Proses konversi ini dilakukan dengan memanfaatkan struktur data Octree, yaitu sebuah tree yang setiap percabangan terdiri dari 8 anak. Melalui Octree, sebuah ruang kubus 3D dapat dipotong menjadi 8 kubus berbeda, yang masing-masing dapat dipotong lagi menjadi 8 kubus yang lebih kecil.

    Tugas Anda adalah mengimplementasikan sebuah program yang dapat mengkonversi model .obj yang umum menjadi model .obj yang terdiri dari voxel dengan memanfaatkan properti Octree. Program melakukan konversi model dengan menggunakan algoritma divide and conquer.

3. Langkah-langkah Penyelesaian

    Untuk melakukan *voxelization* pada suatu file .obj, saya menggunakan 3 buah file, yakni *main*, *utils*, dan *solver*. File *utils* berisi definisi tipe data abstrak yang digunakan, deklarasi variabel global, serta fungsi-fungsi dan prosedur-prosedur untuk pembacaan dan penulisan file atau kegunaan-kegunaan praktis lainnya. File *solver* berisi fungsi-fungsi utama dalam penyelesaian menggunakan *divide and conquer*. Terakhir, file *main* berisi program utama yang menggunakan fungsi-fungsi dan prosedur-prosedur dari *utils* dan *solver* untuk menyelesaikan permasalahan (melakukan *voxelization*).

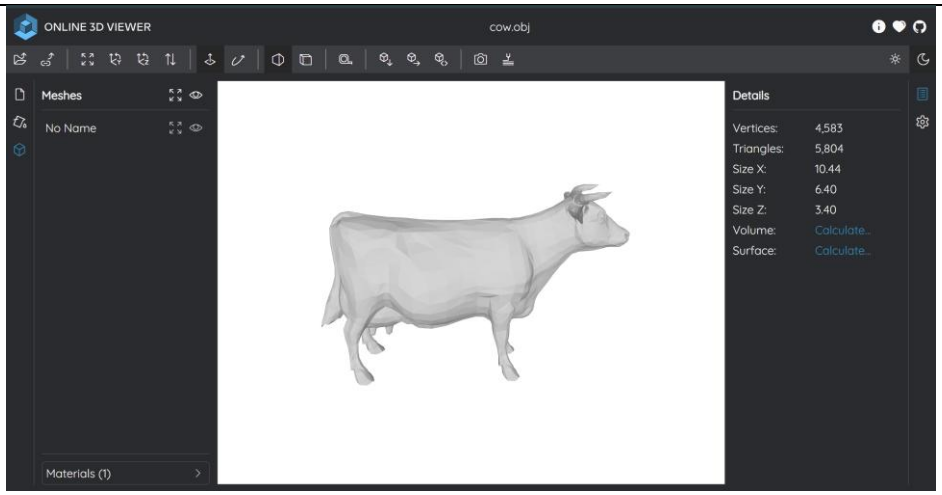
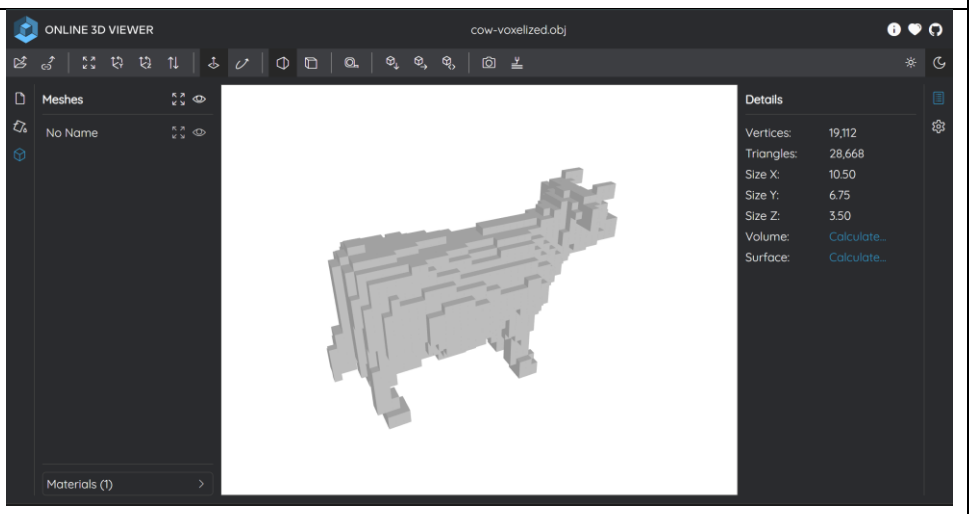
    Strategi *divide and conquer* yang digunakan untuk menyelesaikan permasalahan ini adalah sebagai berikut:

    - Pertama, program membaca file masukan kemudian menyimpan *faces* dan *vertexes* masing-masing pada sebuah vector.
    - Selanjutnya, program akan mencari nilai ekstrem dari koordinat pada *vertexes* yang ada untuk dijadikan acuan ukuran octree root.

- Program akan membagi setiap kubus menjadi 8 bagian, sebanyak derajat octree yang diinput user.
- Apabila kubus yang sedang dirujuk sekarang tidak berpotongan pada *face* atau *vertex* mana pun, maka penelusuran langsung dihentikan (walaupun belum sampai derajat maximum)
- Apabila sudah mencapai derajat maximum octree, apabila kubus yang sedang dirujuk berpotongan dengan sebuah *face* atau *vertex* maka kubus tersebut dibuat dan *vertex* yang menunjukkan titik sudut kubus tersebut serta *face* yang menunjukkan sisi kubus tersebut disimpan masing-masing pada vector *vertex_hasil* dan *face_hasil*.

4. *Test Case*

Test Case 1	
File Input	cow.obj
File Output	cow-voxelized.obj
CLI Output	SELAMAT DATANG DI PROGRAM VOXELISASI POLYGON Masukkan nama file .obj: cow Masukkan kedalaman octree : 7 PROCESSING... Filename : cow.obj Max Depth : 7 Voxel berhasil dibuat! ========== VOXELIZATION STATISTICS ========== Voxel Count : 2389 Vertex Count : 19112 Face Count : 28668 Octree Built (depth= = 1) : 2389 Octree Built (depth= = 2) : 615 Octree Built (depth= = 3) : 163 Octree Built (depth= = 4) : 35 Octree Built (depth= = 5) : 12 Octree Built (depth= = 6) : 8 Octree Built (depth= = 7) : 1 Octree Skipped (depth= = 1) : 259755 Octree Skipped (depth= = 2) : 32153 Octree Skipped (depth= = 3) : 3933 Octree Skipped (depth= = 4) : 477 Octree Skipped (depth= = 5) : 52 Octree Skipped (depth= = 6) : 0 Octree Skipped (depth= = 7) : 0 Max Octree Depth : 7 Duration : 6290 ms Successfully saved result at test/output/cow-voxelized.obj ==========

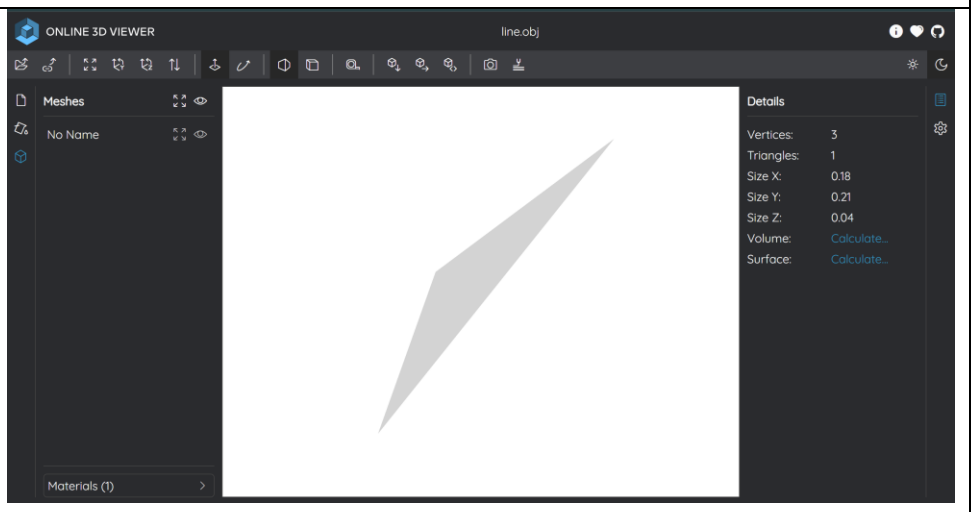
Visualisasi File Input	
Visualisasi File Output	

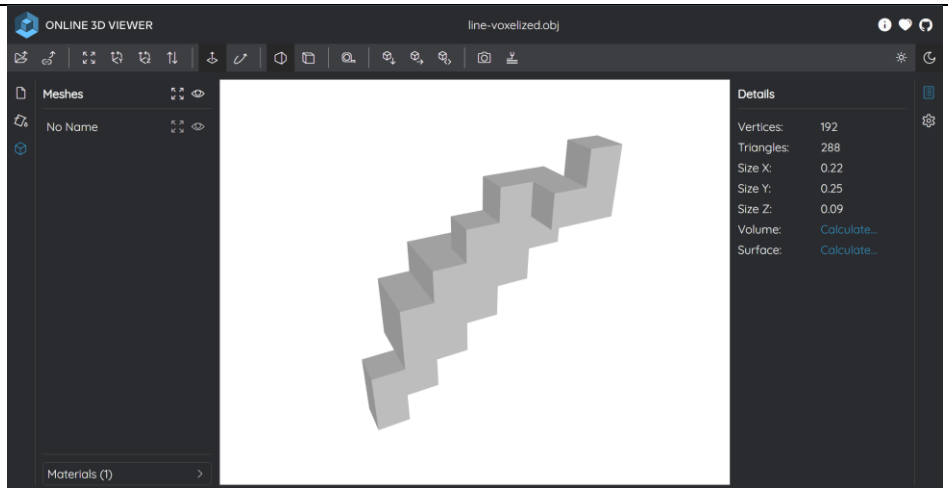
Test Case 2	
File Input	line.obj
File Output	line-voxelized.obj

CLI Output	

```
SELAMAT DATANG DI PROGRAM VOXELISASI POLYGON
Masukkan nama file .obj: line
Masukkan kedalaman octree : 9
PROCESSING...
Filename : line.obj
Max Depth : 9
Voxel berhasil dibuat!
==================== VOXELIZATION STATISTICS ====================
Voxel Count                 : 24
Vertex Count                : 192
Face Count                  : 288
Octree Built (depth= = 1)   : 24
Octree Built (depth= = 2)   : 12
Octree Built (depth= = 3)   : 6
Octree Built (depth= = 4)   : 2
Octree Built (depth= = 5)   : 1
Octree Built (depth= = 6)   : 1
Octree Built (depth= = 7)   : 1
Octree Built (depth= = 8)   : 1
Octree Built (depth= = 9)   : 1
Octree Skipped (depth= = 1) : 16777192
Octree Skipped (depth= = 2) : 2097140
Octree Skipped (depth= = 3) : 262138
Octree Skipped (depth= = 4) : 32766
Octree Skipped (depth= = 5) : 4095
Octree Skipped (depth= = 6) : 511
Octree Skipped (depth= = 7) : 63
Octree Skipped (depth= = 8) : 7
Octree Skipped (depth= = 9) : 0
Max Octree Depth            : 9
Duration                    : 1 ms
Successfully saved result at test/output/line-voxelized.obj
====================================================================
```

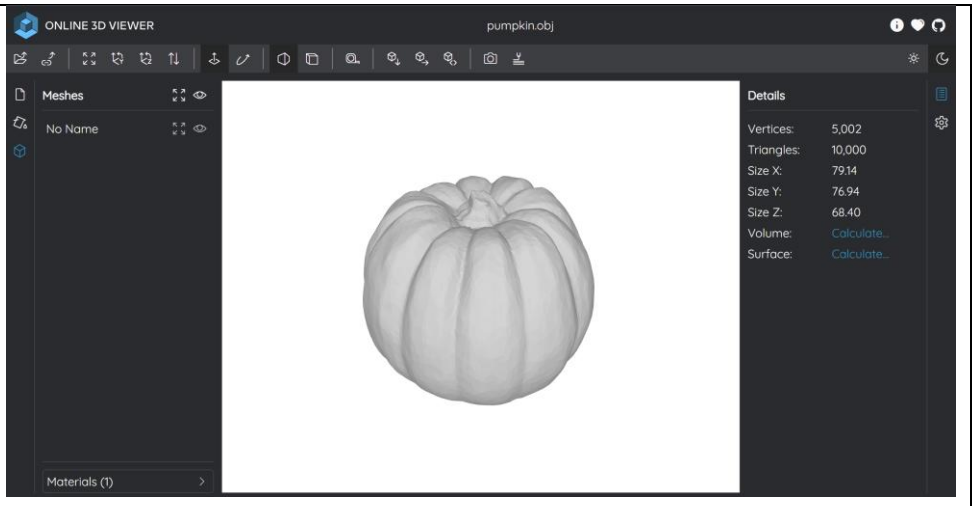
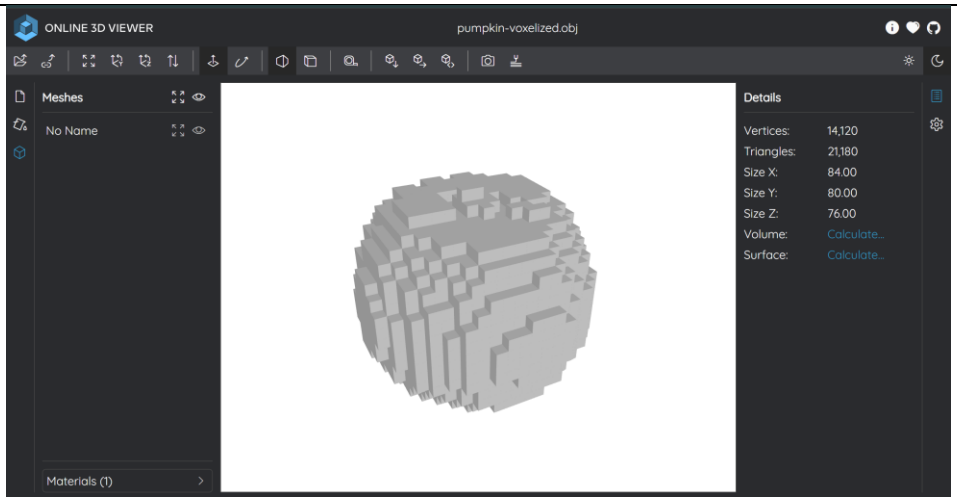
Visualisasi File Input



Visualisasi File Output	

Test Case 3	
File Input	pumpkin.obj
File Output	pumpkin-voxelized.obj
CLI Output	

```
SELAMAT DATANG DI PROGRAM VOXELISASI POLYGON
Masukkan nama file .obj: pumpkin
Masukkan kedalaman octree : 8
PROCESSING...
Filename : pumpkin.obj
Max Depth : 8
Voxel berhasil dibuat!
==================== VOXELIZATION STATISTICS ====================
Voxel Count                 : 1765
Vertex Count                : 14120
Face Count                  : 21180
Octree Built (depth= = 1)   : 1765
Octree Built (depth= = 2)   : 439
Octree Built (depth= = 3)   : 116
Octree Built (depth= = 4)   : 36
Octree Built (depth= = 5)   : 8
Octree Built (depth= = 6)   : 8
Octree Built (depth= = 7)   : 4
Octree Built (depth= = 8)   : 1
Octree Skipped (depth= = 1) : 2095387
Octree Skipped (depth= = 2) : 261705
Octree Skipped (depth= = 3) : 32652
Octree Skipped (depth= = 4) : 4060
Octree Skipped (depth= = 5) : 504
Octree Skipped (depth= = 6) : 56
Octree Skipped (depth= = 7) : 4
Octree Skipped (depth= = 8) : 0
Max Octree Depth            : 8
Duration                    : 7495 ms
Successfully saved result at test/output/pumpkin-voxelized.obj
====================================================================
```

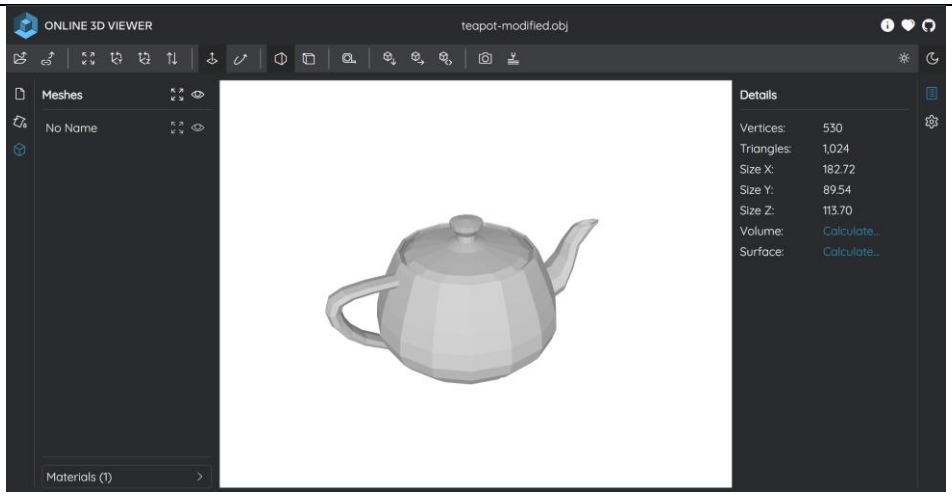
Visualisasi File Input	
Visualisasi File Output	

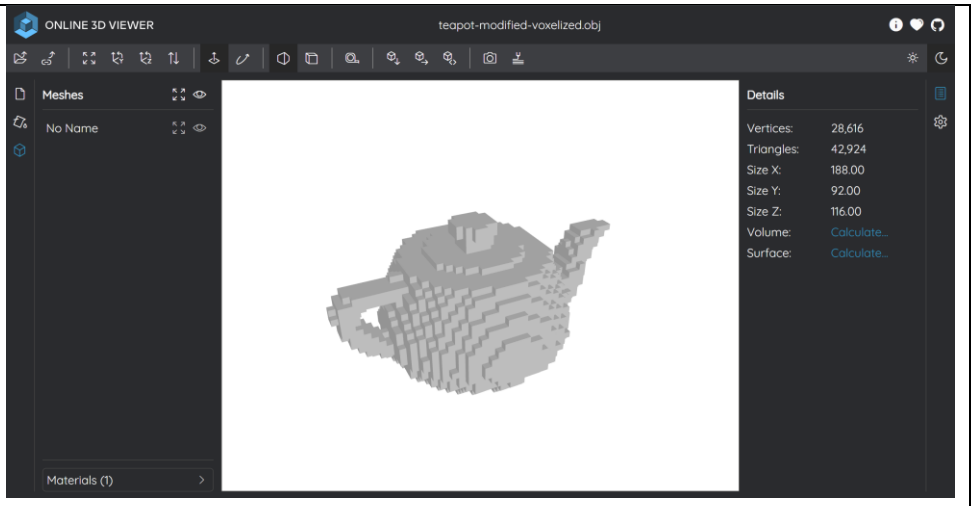
Test Case 4	
File Input	teapot-modified.obj
File Output	teapot-modified-voxelized.obj

CLI Output	

```
SELAMAT DATANG DI PROGRAM VOXELISASI POLYGON
Masukkan nama file .obj: teapot-modified
Masukkan kedalaman octree : 7
PROCESSING...
Filename : teapot-modified.obj
Max Depth : 7
Voxel berhasil dibuat!
==================== VOXELIZATION STATISTICS ====================
Voxel Count                 : 3577
Vertex Count                : 28616
Face Count                  : 42924
Octree Built (depth= = 1)   : 3577
Octree Built (depth= = 2)   : 855
Octree Built (depth= = 3)   : 221
Octree Built (depth= = 4)   : 58
Octree Built (depth= = 5)   : 16
Octree Built (depth= = 6)   : 8
Octree Built (depth= = 7)   : 1
Octree Skipped (depth= = 1) : 258567
Octree Skipped (depth= = 2) : 31913
Octree Skipped (depth= = 3) : 3875
Octree Skipped (depth= = 4) : 454
Octree Skipped (depth= = 5) : 48
Octree Skipped (depth= = 6) : 0
Octree Skipped (depth= = 7) : 0
Max Octree Depth            : 7
Duration                    : 1822 ms
Successfully saved result at test/output/teapot-modified-voxelized.obj
==================================================================
```

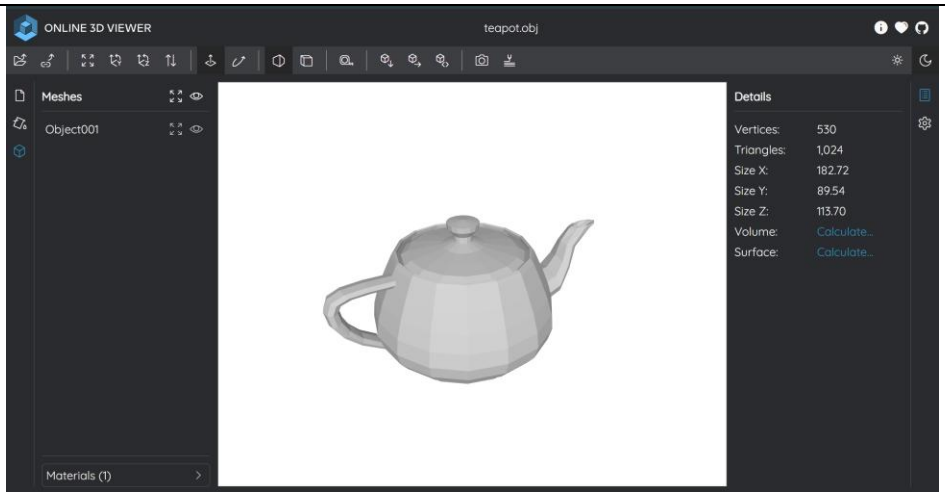
Visualisasi File Input



Visualisasi File Output	

Test Case 5	
File Input	teapot.obj
File Output	-

CLI Output	`SELAMAT DATANG DI PROGRAM VOXELISASI POLYGON`
	`Masukkan nama file .obj: teapot`
	`Isi file tidak valid, terdapat baris yang tokennya terlalu banyak`
	`Error line: 1`

Visualisasi File Input	
Visualisasi File Output	-

5. Pralana *Repository*

   https://github.com/MFakhryzaKi/Tucil2_13524087

6. Pernyataan

Tugas ini disusun sepenuhnya tanpa bantuan kecerdasan buatan (Generative AI), melainkan hasil pemikiran dan analisis mandiri.

Muhammad Fakhry Zaki

7. Lampiran

No	Poin	Ya	Tidak
1	Program berhasil di kompilasi tanpa kesalahan	✓	
2	Program berhasil dijalankan	✓	
3	Hasil konversi .obj terdiri dari kubus-kubus kecil	✓	
4	Program informasi-informasi atau report	✓	
5	Program dapat menerima masukan .obj, dan mengembalikan output	✓	
6	[Bonus] Program menggunakan *concurrency*		✓
7	[Bonus] Program dapat menampilkan visualisasi .obj		✓